

Wednesday Reading Assessment: Unit 3, Magnetic Forces and Fields

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1 Memory Bank

- $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$... The direction of the *cross-product* follows this pattern. Reversing the order of any two vectors introduces a minus sign.
- $\hat{i} \times \hat{i} = 0$, $\hat{j} \times \hat{j} = 0$, $\hat{k} \times \hat{k} = 0$... The *cross-product* is zero when both vectors are parallel.
- $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin\theta$... The magnitude of the cross-product of two vectors $\vec{a}$ and $\vec{b}$, given the angle θ between them.
- $\vec{F} = i\vec{L} \times \vec{B}$... The Lorentz force, by a magnetic field $\vec{B}$ on a *current* i of length and direction $\vec{L}$.

2 Voltmeters and Ammeters

1. Suppose the ammeter in Fig. 1 has an internal resistance $r = 10\Omega$, and a “shunt” resistance of $R = 0.1\Omega$. The circuit has $R_1 = 1\text{ k}\Omega$, $R_2 = 1\text{ k}\Omega$, $\mathcal{E} = 12\text{V}$ and $r_b = 1\Omega$. (a) Calculate the total resistance of the ammeter, R_A . (b) Calculate the total current flow from the battery, I_{total} . (c) What is the decrease in voltage across the ammeter? (d) What is the current flow to G, the galvanometer, I_G ?

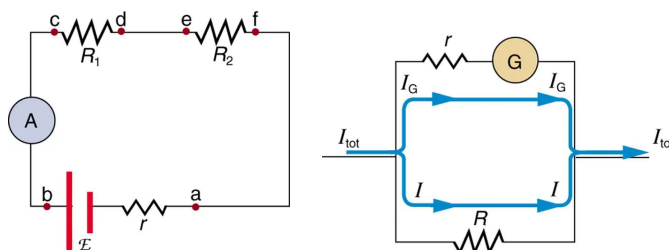


Figure 1: (Left) An ammeter in series with R_1 and R_2 , and a battery with emf $\mathcal{E}$ and internal resistance r_b . (Right) The internal structure of the ammeter, with galvanometer G, internal resistance r and shunt resistance R .

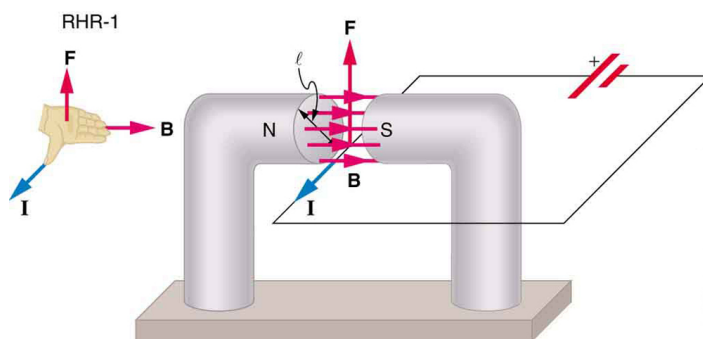


Figure 2: A permanent magnet creates a magnetic field to the *right*, while a voltage pushes a current *out of the page*. The force is measured to occur in the *upward* direction.

3 The Cross-Product

1. Let $\vec{a} = 4\hat{i}$, and $\vec{b} = 4\hat{j}$. (a) Calculate $\vec{a} \times \vec{b}$. (b) What are the magnitude and direction of the result?
2. What is the angle between the two vectors in the previous problem?

4 Magnetic Force on a Wire

1. Consider Fig. 2, in which a current passes through a *magnetic field* generated by a permanent magnet. (a) If the length of wire in the B-field is $L = 10\text{ cm}$, the magnetic field is $B = 10^{-1}\text{ Tesla}$, the voltage is $V = 24\text{ Volts}$, and the effective resistance in the wire is $R = 2\Omega$, what is the force upwards on the wire? (b) If the wire was attached to a scale, and the scale has a mass of $m = 24\text{ grams}$ on it, what would the scale read if we turn on the current?¹

¹Hint: it won't say 24 grams.